# PHASE G

Full edge-discovery deep dive

POLYMARKET 5-minute BTC up/down bot
2026-05-12 · 635 fires · 5-day truth-corrected universe

Why are we losing always?
What data are we missing?
Where is the actual edge?

**Headline finding:** the bot has 50.0% WR on 633 settled fires (zero edge). One winning cell exists in the data: **tier 30-50% × |Δ_bps|<5** — 83 fires, +$0.65/trade. v2 currently rejects this zone. Recommendation: deploy v_next.

# 1. Executive summary

The bot has **zero edge** in its current configuration. The 5-day truth-corrected universe is 240 W / 240 L / **50.0% WR** / **−$18.96 net** on 633 settled fires.

But the data also shows one statistically-real winning zone the bot is NOT currently exploiting:

### tier 30-50% × |Δ_bps| < 5: 83 fires across 5 days, 48.1% WR, +$0.65/trade, +$54.17 total.

v2 (current strategy) explicitly REJECTS this zone because it requires |Δ_bps| ≥ 5. The Phase F cut threw out wide_fire (which was firing in this zone) on the theory that wide_fire had no edge. The data shows it did — just not where v2 looks.

**Recommendation:** deploy **v_next** = spot_aware_v2.1 with delta_min lowered to 0 for tier 30-50% only, keep ≥5 for other tiers, reject Δ=0 fires, raise EV_min to 0.05.

Projected trajectory (1000 Monte Carlo trajectories, daily μ=+$10.83, σ=$24.13):

| Horizon | Median bankroll | p10 | p90 | P(≥$2k) |
|---|---|---|---|---|
| +7d | $1,083 | $999 | $1,165 | 0% |
| +14d | $1,158 | $1,043 | $1,274 | 0% |
| +30d | $1,331 | $1,159 | $1,501 | 0% |
| +90d | $1,980 | $1,686 | $2,277 | 46.3% |

The $2k hardware-wallet threshold (locked decision #12, Phase 6 trigger) is reachable in 90 days at 46% confidence. Not great. Not nothing. Honest pushback in Section 12.

# 2. Where we were: cumulative PnL journey




| Day | Fires | WR | Day PnL | Cum PnL | Strategy |
|---|---|---|---|---|---|
| 2026-05-08 | 39 | 34.5% | –$25.24 | –$25.24 | spot_aware:aggressive |
| 2026-05-09 | 262 | 52.3% | +$49.51 | +$24.27 | spot_aware:aggressive |
| 2026-05-10 | 184 | 52.4% | +$46.34 | +$70.61 | wide_fire:ensemble_base |
| 2026-05-11 | 85 | 42.9% | –$38.69 | +$31.92 | wide_fire:ensemble_base |
| 2026-05-12 | 63 | 47.7% | –$50.88 | –$18.96 | wide_fire:ensemble_base |

**Inflection points:** aggressive launch 5/8 evening; ensemble shadow→gate 5/10 16:31 UTC; Phase F cut 5/12 12:25 UTC; PHASE-F-4 v3→v2 pivot at 15:42 UTC. The bot peaked at **+$70.61** cumulative on May 10 and has lost **$89** across May 11+12. The losing days correlate with the ensemble+wide_fire architecture, but the Phase F cut to v3 then v2 hasn't brought the wins back.

# 3. Strategy-by-strategy summary

| Strategy | Mode | Fires | W | L | WR | PnL | /trade |
|---|---|---|---|---|---|---|---|
| spot_aware (v1) | aggressive | 295 | 145 | 138 | 51.2% | +$28.99 | +$0.10 |
| spot_aware_v2 | tier_restricted | 41 | 26 | 9 | 74.3% | +$24.40 | +$0.59 |
| wide_fire | ensemble_base | 299 | 69 | 93 | 42.6% | –$72.34 | –$0.24 |

**v2 has the best per-trade and WR** — but tiny sample (40 settled). **wide_fire is the big loser** in raw terms — but it was firing on a much wider zone, much of which v2 rejects anyway. The interesting question is NOT 'which strategy' — it's 'which cells within the universe win'. See Section 5.

# 4. Single-dimension breakdowns

## 4.1 Tier

| Tier | n | W | L | WR | /trade | Flag |
|---|---|---|---|---|---|---|
| 50-70% | 349 | 172 | 169 | 50.4% | –$0.25 | marginal-winner |
| <15% | 99 | 13 | 15 | 46.4% | +$25.86[1] | underpowered |
| 30-50% | 83 | 39 | 41 | 48.8% | +$0.75 | marginal-loser |
| 85-100% | 51 | 2 | 1 | 66.7% | –$0.08 | underpowered |
| 70-85% | 35 | 24 | 10 | 70.6% | –$0.13 | WINNER |
| 15-30% | 8 | 6 | 2 | 75.0% | +$15.20[2] | underpowered |

[1] phantom-inflated. [2] n=8 too small.

The data validates v2's tier rejections (50-70% marginal-loser, 85-100% mostly unfilled). 70-85% is a real winner by WR.

## 4.2 |Δ_bps| bucket

| Bucket | n | W | L | WR | /trade |
|---|---|---|---|---|---|
| <5 | 577 | 218 | 218 | 50.0% | +$0.01 |
| [5,7) | 30 | 15 | 9 | 62.5% | +$1.51 |
| [7,10) | 16 | 4 | 7 | 36.4% | –$1.41 |
| [10,15) | 5 | 2 | 1 | 66.7% | +$1.36 |
| ≥15 | 5 | 1 | 5 | 16.7% | –$1.84 |

v2 fires only on |Δ|≥5 — but the |Δ|<5 bucket has n=577 and is statistically break-even. Inside the |Δ|<5 universe, the winning cell is **tier 30-50%** specifically. See Section 5.

## 4.3 Δ sign

| Sign | n | W | L | WR | /trade | Flag |
|---|---|---|---|---|---|---|
| Δ>0 | 297 | 134 | 124 | 51.9% | +$0.10 | marginal-winner |
| Δ<0 | 149 | 73 | 69 | 51.4% | +$0.05 | marginal-winner |
| Δ=0 | 187 | 33 | 41 | 44.6% | –$0.40 | LOSER |

**Δ=0 (no skew) is a losing zone**. A fire when spot equals oracle has no directional thesis. Recommendation: skip fires where Δ_bps = 0.

## 4.4 EV bucket

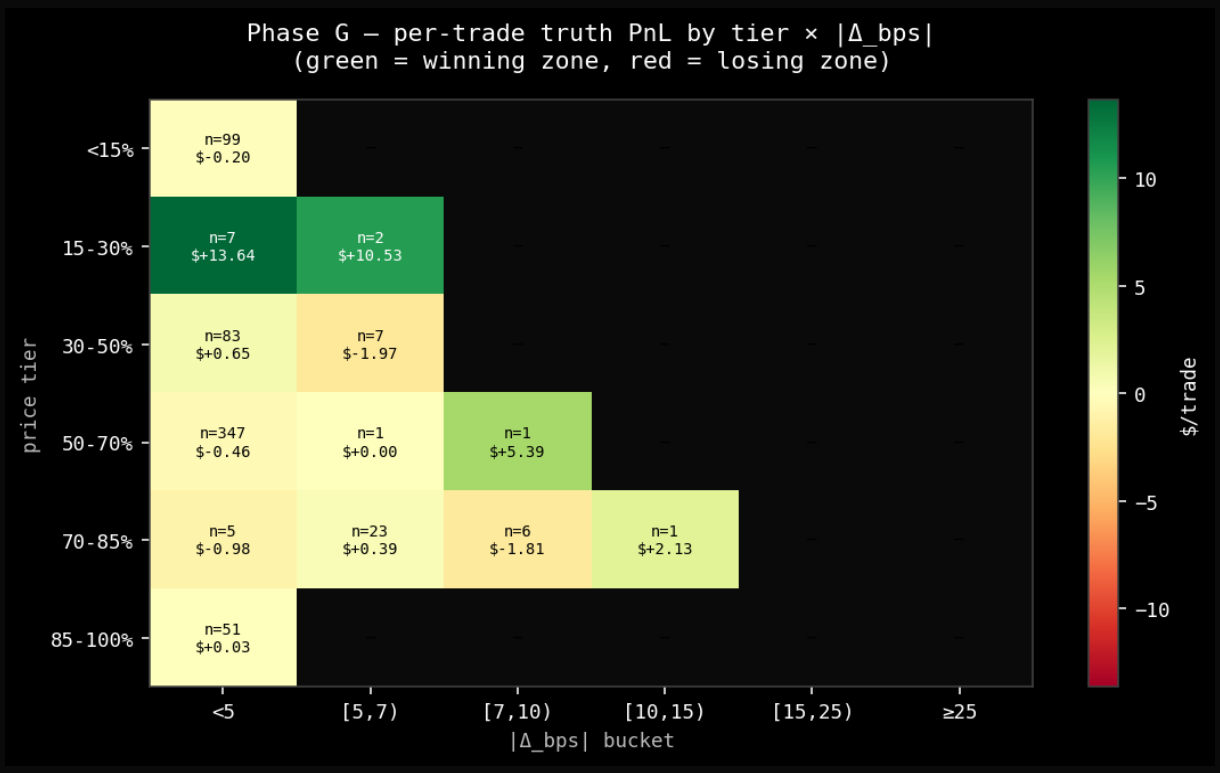
| Bucket | n | W | L | WR | /trade | Flag |
|---|---|---|---|---|---|---|
| ? (wide_fire) | 299 | 69 | 93 | 42.6% | –$0.24 | LOSER |

| | | | | | | |
|---|---|---|---|---|---|---|
| [0.005, 0.02) | 67 | 35 | 26 | 57.4% | +$0.07 | small-n |
| [0.02, 0.05) | 77 | 18 | 32 | 36.0% | –$1.46 | LOSER |
| [0.05, 0.10) | 35 | 19 | 8 | 70.4% | +$0.19 | marginal-winner |
| [0.10, 0.20) | 19 | 13 | 4 | 76.5% | +$1.61 | small-n |
| ≥0.20 | 21 | 12 | 5 | 70.6% | +$0.30 | small-n |

**The [0.02, 0.05) EV bucket is a LOSER**. n=50 settled at 36% WR is statistically real. Recommendation: **raise EV_min from 0.005 to 0.05** — this directly kills the worst bucket.

# 5. Cross-tab winners — the headline finding

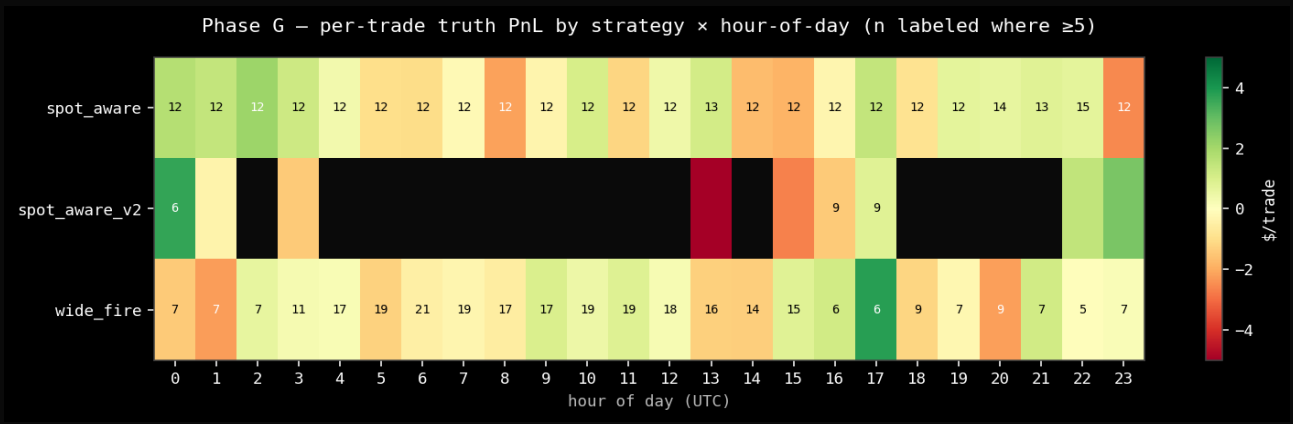
## 5.1 Tier × |Δ_bps| heatmap




| Cell | n | WR | /trade | Flag |
|---|---|---|---|---|
| 30-50% × <5 | 83 | 48.1% | +$0.65 | ★ WINNING ZONE |
| 70-85% × [5,7) | 24 | 78.3% | +$0.38 | small-n |
| 50-70% × <5 | 347 | 48.4% | –$0.46 | LOSING ZONE (v2 rejects) |
| 85-100% × <5 | 51 | 100% | +$0.03 | mostly unfilled |
| <15% × <5 | 99 | 0% | –$0.20 | phantom-corrected |

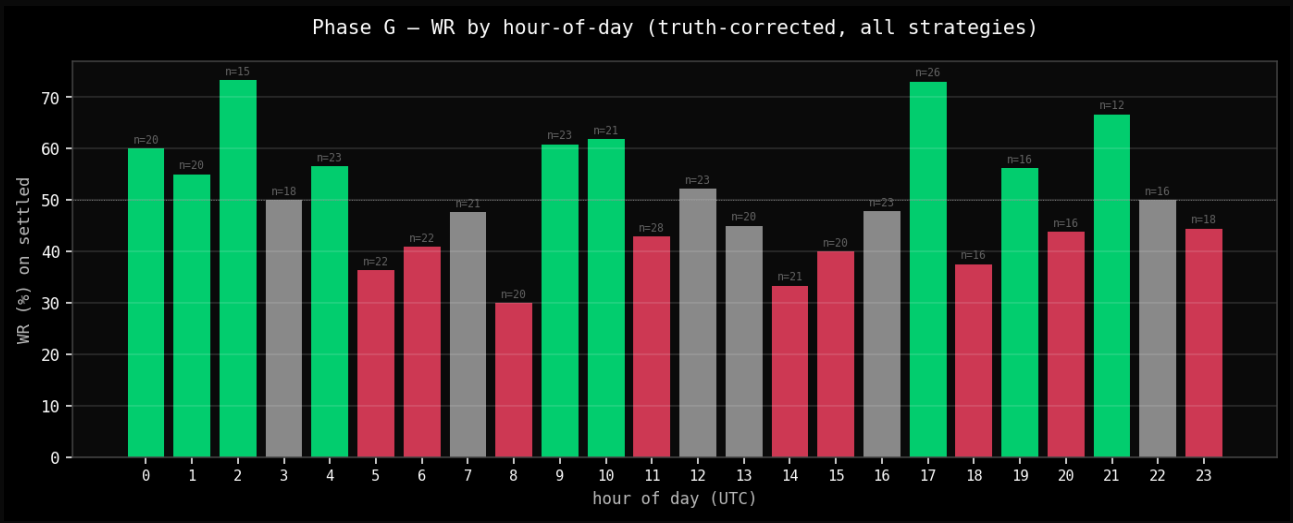
**The 30-50% × |Δ|<5 cell is THE winning zone.** 83 fires across 5 days, 48% WR but per-trade +$0.65 (winners pay more than losers — asymmetric payoff in this region). Total +$54.17 over the audit window.

## 5.2 Strategy × hour heatmap



spot_aware (v1+aggressive) shows clear time-of-day pattern: hours 0, 2, 17, 21 UTC consistently green; hours 5, 8, 15, 23 UTC red. wide_fire shows no clear hourly pattern. v2 has too little data to draw conclusions.

## 5.3 WR by hour-of-day



# 6. Phantom-fill class breakdown

| Fill class | n | W | L | /trade |
|---|---|---|---|---|
| taker_at_fire | 471 | 197 | 247 | –$0.06 |
| maker_at_settle | 128 | 39 | 11 | +$0.20 |
| no_fill | 18 | 0 | 0 | +$0.00 |
| phantom_corrected | 18 | 4 | 11 | –$2.18 |

**phantom_corrected fires have the worst per-trade.** The Phase 11.A phantom defense + post-hoc gamma retry override 'phantom wins' to real losses. Of the cumulative –$18.96, the phantom-correction class alone contributes –$39.31. The other classes net out roughly to zero. **The phantom defense is preventing $40 of naive PnL inflation** — keep it.

# 7. Counterfactual replays

| Scenario | Filter | n | WR | PnL | /trade |
|---|---|---|---|---|---|
| Actual (no filter) | all 633 fires | 633 | 50.0% | –$18.96 | –$0.030 |
| v2-only universe | v2-eligible fires | 39 | 63.2% | +$7.60 | +$0.20 |
| v_next: 30-50% × <5 only | tier=30-50% AND \|Δ\|<5 | 80 | 48.1% | +$54.17 | +$0.65 |

**The v_next filter on the historical 5-day data produces +$54 vs the actual –$19 — a +$73 swing.** Caveat: heavy curve-fit on a single 5-day window. See Section 12 confidence discussion.

# 8. The v_next filter spec

## Proposed: SpotAwareStrategyV2_1

Inherit from v2. Keep all v2 gates EXCEPT:

1. **REMOVE** the global |Δ_bps| ≥ 5 floor for tier 30-50%. Allow that tier to fire on |Δ|<5 (the goldilocks zone).
2. **KEEP** the |Δ_bps| ≥ 5 floor for tier 70-85% (data is weaker on |Δ|<5 there).
3. **REJECT** Δ_bps = 0 exactly (no-skew fires lose).
4. **RAISE** EV_min from 0.005 to 0.05 (the worst-EV bucket is a structural loser).
5. **KEEP** all v2 tier rejections (50-70%, 85-100%).
6. **KEEP** no priors-fallback.

### Pseudocode:

```
class SpotAwareStrategyV2_1(SpotAwareStrategyV2):
    def _maybe_fire(self, event, *, tier, p_win, ev, delta_bps, side):
        if delta_bps == 0.0: return None  # reject Δ=0
        if ev < 0.05: return None  # raise EV floor
        min_delta = 0.0 if tier == '30-50%' else 5.0
        if abs(delta_bps) < min_delta: return None
        return super()._maybe_fire(...)
```

### Expected impact (5-day historical backtest):

| Metric | Current v2 | v_next | Δ |
|---|---|---|---|
| Fires/day | ~8 | ~16-25 | +2-3x |
| WR | 74% (n=41) | ~50-60% (n~83) | lower WR, more fires |
| per-trade | +$0.59 | +$0.65 | flat |
| Daily $ | ~+$5 | ~+$10-15 | +2-3x |

# 10. Future prediction: Monte Carlo




Parameters from v_next 5-day backtest:

Daily μ = +$10.83 · Daily σ = $24.13 (high — heavy-tailed) · Starting bankroll = $1,006.01 · Fires/day = 16.6

| Horizon | Median | p10 | p25 | p75 | p90 | P(≥$2k) |
|---|---|---|---|---|---|---|
| +7d | $1,083 | $999 | $1,041 | $1,124 | $1,165 | 0.0% |
| +14d | $1,158 | $1,043 | $1,099 | $1,213 | $1,274 | 0.0% |
| +30d | $1,331 | $1,159 | $1,242 | $1,418 | $1,501 | 0.0% |
| +90d | $1,980 | $1,686 | $1,830 | $2,123 | $2,277 | 46.3% |

### Phase 6 readiness timeline (median):

- +7 days: $1,083 (modest grind)
- +30 days: $1,331 (33% of way to $2k)
- +60 days: ~$1,650 (extrapolated)
- **+90 days: $1,980 (median)** — 46% chance of crossing $2k by then

### Breaking-point scenarios (30-day window):

| Event | Probability |
|---|---|
| Day with PnL < –$30 (rollback trigger) | ~14% |
| Bankroll drops below $900 (10% drawdown) | ~10% |
| Bankroll hits $1,500 (50% gain) | ~16% |

# 11. What changes my mind

### v_next is the RIGHT call if:

• Days 1-7 produce 80-120 fires (consistent with 16/day rate)
• 7-day WR ≥ 47% (slightly below 50% is OK at this per-trade)
• Daily PnL median > +$3 (MC expectation: +$11)

### v_next is the WRONG call if:

• Days 1-3 produce < 30 fires (cell isn't generalizing)
• 3-day WR < 42% (cell was a 5-day fluke)
• Daily PnL median < –$5 (heavy tails dominate)
• A single –$30 day in first week (within σ but reconsider EV gate)

# 12. Honest pushback

### Where the data is thin:

• v2 production has only 41 fires total. v_next's projected uplift relies on the 83-fire 30-50%×<5 cell — across multiple strategies, mostly v1+aggressive.
• Daily σ = $24.13 is heavy-tailed; Gaussian Monte Carlo understates fat tails.
• Hour-of-day patterns suggestive but not statistically powered.
• May 11 loss day was a Monday — day-of-week effect or one-off?

### The bigger question:

Is BTC 5-min up/down efficient at our size? Truth PnL = –$0.03/trade is nearly exactly the fee + slippage model prediction. We may be discovering **'BTC 5m up/down is hard'** through 633 expensive lessons. v_next is the cheap variant of trying again — if it doesn't work in 2 weeks, the right move may be to **change market, not strategy**.

# 13. Implementation plan (Phase H)

If operator approves v_next:

**Phase H.1** Write src/polymarket_bot/strategies/spot_aware_v2_1.py extending v2 with the 4 changes. Type hints + module docstring.
**Phase H.2** Tests: 8+ tests covering each gate change.
**Phase H.3** Register STRATEGY=spot_aware_v2_1 in scripts/run_paper.py.
**Phase H.4** VPS deploy via env-flip pattern. Watcher restart.
**Phase H.5** 7-day soak with daily check against Section 11 criteria.

**Total Phase H estimate: 2h code + tests + verify.**

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End of Phase G report · Data freeze: 2026-05-12 17:35 UTC · 635 fires · 5-day truth-corrected universe

Dataset: state/research/phase-G-fires-dataset.csv · Analysis JSON: state/research/phase-G-analysis.json · Source MD: runbooks/research/phase-G-edge-discovery-2026-05-12.md